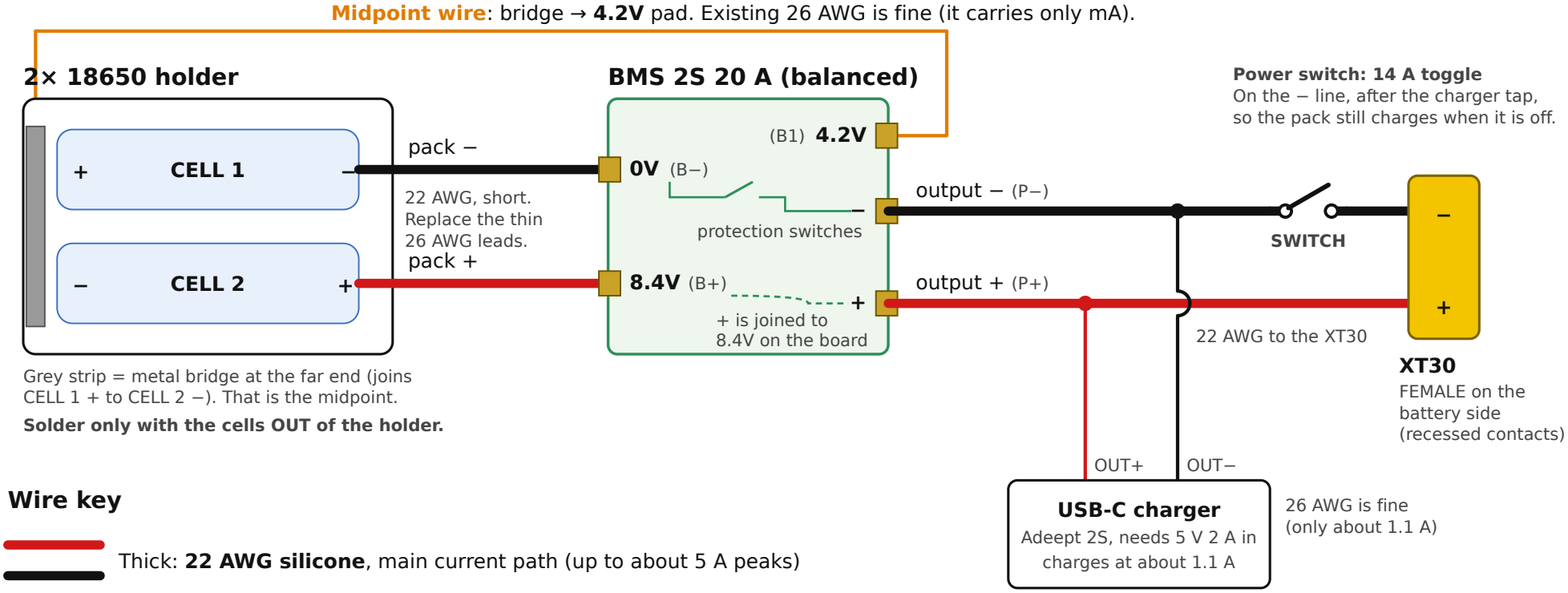
Open Duck Mini v2: battery pack wiring

For your parts: 2× 18650 holder · 2S 20 A balanced BMS (pads 0V / 4.2V / 8.4V / − / +) · Adeept 2S USB-C charger · XT30

Pads are drawn on the edges you described. Their order along each edge may differ on your board, so go by the printed labels, not by position.



Wire key

Thick: **22 AWG silicone**, main current path (up to about 5 A peaks)

Thin: 26 AWG is fine (charger, midpoint)

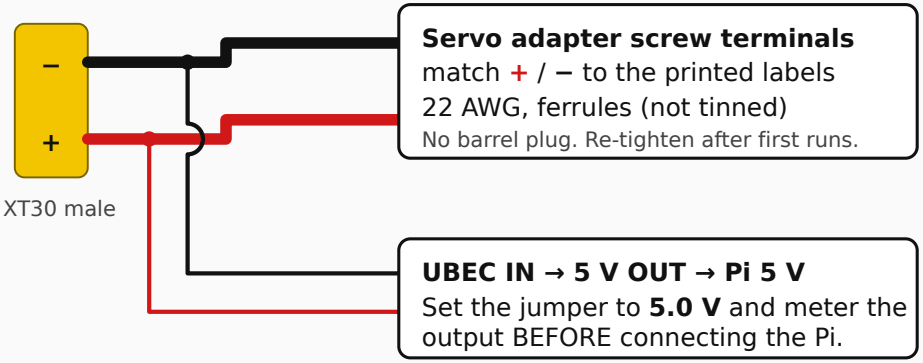
Red = positive · Black = negative · **Orange** = midpoint (balance)

Before the cells go in (meter on Ω / continuity)

- ☐ **+** to **8.4V**: about 0 Ω (same copper)
- ☐ **−** to **0V**: **NOT** 0 Ω (protection switches in between)
- ☐ 4.2V to 0V and 4.2V to 8.4V: not shorted
- ☐ Load and charger negative go to **−**, never to 0V

After the cells go in (DC volts). Yours: 3.720 / 3.721 V

- ☐ 0V→4.2V ≈ 3.72 V · 4.2V→8.4V ≈ 3.72 V · 0V→8.4V ≈ 7.44 V
- ☐ − to + ≈ 7.44 V. Reads 0 V? Plug in the charger for a few seconds to wake the BMS.
- ☐ First charge: watch it, never leave it unattended. It stops at about 8.4 V, halves within 0.05 V, cells stay cool.
- ☐ **Don't charge to full yet:** the servos' limit is 8.0 V. Use the pack as it is (7.44 V) for the first tests.

Robot side: XT30 to servo adapter and UBEC

Splits two ways: servo adapter (heavy current) and UBEC (under 1 A).